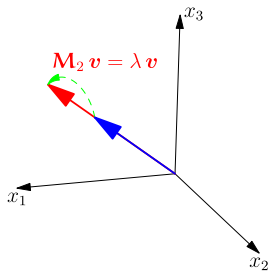


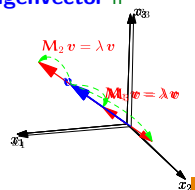
Eigensystems

Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank

Eigenvector equation

- Eigen-systems help us to understand mappings
- A non-zero vector v is said to be an **eigenvector** if

$$Mv = \lambda v$$



- M is square (i.e. $n \times n$)
- Where the number λ is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

Proof of Orthogonality

- $(Mv_i = \lambda_i v_i)^T$ implies $v_i^T M^T = \lambda_i v_i^T$
- When M is symmetric then $Mv_i = \lambda_i v_i \Rightarrow v_i^T M = \lambda_i v_i^T$
- Consider two eigenvectors v_i and v_j of M

$$\begin{aligned} v_i^T M v_j &= (v_i^T M) v_j = \lambda_i v_i^T v_j \\ &= v_i^T (M v_j) = \lambda_j v_i^T v_j \end{aligned}$$

- So either $\lambda_i = \lambda_j$ or $v_i^T v_j = 0$
- If $\lambda_i = \lambda_j$ then any linear combination of v_i and v_j is an eigenvector ($M(av_i + bv_j) = \lambda_i(av_i + bv_j)$). So I can choose two eigenvectors that are orthogonal to each other.

Orthogonal Matrices

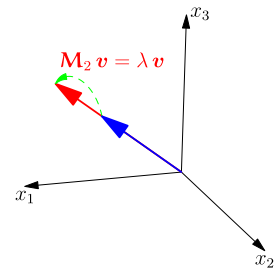
- We can construct an **orthogonal** matrix V from the normalised eigenvectors ($v_i^T v_i = 1$)

$$V = (v_1, v_2, \dots, v_n)$$

- Matrix V is an $n \times n$ matrix
- Because the vectors v_i are orthogonal and normalised

$$V^T V = \begin{pmatrix} v_1^T v_1 & v_1^T v_2 & \dots & v_1^T v_n \\ v_2^T v_1 & v_2^T v_2 & \dots & v_2^T v_n \\ \vdots & \vdots & \ddots & \vdots \\ v_n^T v_1 & v_n^T v_2 & \dots & v_n^T v_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = I$$

- Eigenvectors**
- Orthogonal Matrices
- Eigen Decomposition
- Low Rank Approximation
- Non-Normal Matrices



Symmetric Matrices

- If M is an $n \times n$ **symmetric** matrix then it has n real orthogonal eigenvectors with real eigenvalues
- We denote the i^{th} eigenvector by v_i and the corresponding eigenvalue by λ_i so that

$$Mv_i = \lambda_i v_i$$

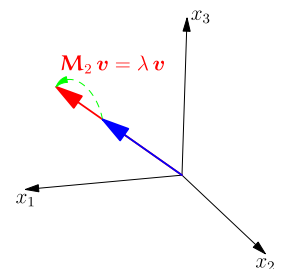
- Orthogonal means that if $i \neq j$ then

$$v_i^T v_j = 0$$

- (We can always normalise eigenvectors if we want)

Outline

- Eigenvectors**
- Orthogonal Matrices**
- Eigen Decomposition
- Low Rank Approximation
- Non-Normal Matrices



The Other Way Around

- We have shown that $V^T V = I$
- Thus multiply both sides on the left by V

$$V V^T V = V$$

- V will have an inverse (we show this on the next slide), V^{-1} , such that $V V^{-1} = I$

- Multiplying the equation on the right by V^{-1}

$$\begin{aligned} (V V^T) V V^{-1} &= V V^{-1} \\ V V^T &= I \end{aligned}$$

- Note that, $V^{-1} = V^T$ (definition of orthogonal matrix)

Invertible Matrices

- A matrix, M , will be singular (not invertible) if there exists a vector $x \neq 0$ such that

$$Mx = 0$$

- Now if there exists such a vector such that $Vx = 0$ then multiplying by V^T we get

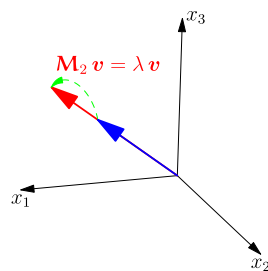
$$\begin{aligned} V^T Vx &= V^T 0 \\ x &= 0 \end{aligned}$$

since $V^T V = I$

- Thus V is invertible

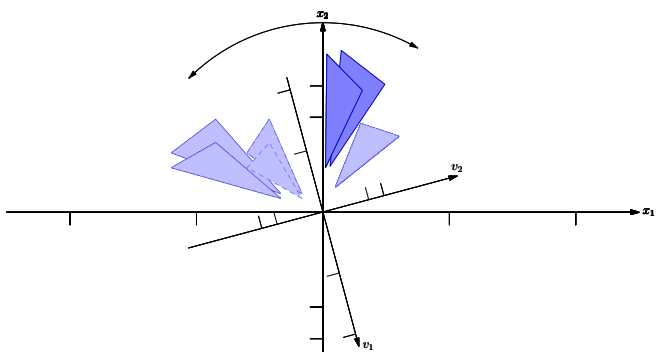
Outline

- Eigenvectors
- Orthogonal Matrices
- Eigen Decomposition**
- Low Rank Approximation
- Non-Normal Matrices

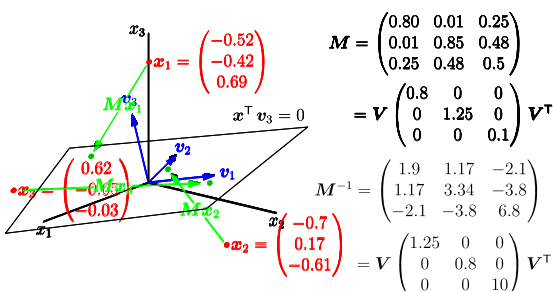


Mappings by Symmetric Matrices

$$M = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = V\Lambda V^T = \begin{pmatrix} \cos(-75^\circ) & -\sin(-75^\circ) \\ \sin(-75^\circ) & \cos(-75^\circ) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75^\circ) & -\sin(75^\circ) \\ \sin(75^\circ) & \cos(75^\circ) \end{pmatrix}$$



III-Conditioning Again



Rotations

- Orthogonal matrices satisfy $V^T V = V V^T = I$
- As a consequence they define rotations (and possibly a reflection)
- Consider a vector x and $x' = Vx$, now

$$\|x'\|_2^2 = x'^T x' = (Vx)^T (Vx) = x^T V^T V x = x^T x = \|x\|_2^2$$

- Similarly if additionally $y' = Vy$ then

$$\langle x', y' \rangle = (Vx)^T (Vy) = x^T V^T V y = x^T y = \langle x, y \rangle = \|x\|_2 \|y\|_2 \cos(\theta)$$

- Rotations and reflections preserve lengths and angles

Matrix Decomposition

- Taking the matrix of eigenvectors, V , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

- Now

$$M = MVV^T = V\Lambda V^T$$

- Very important *similarity transform*

Inverses

- For any symmetric invertible matrix

$$M = V\Lambda V^T \quad M^{-1} = V\Lambda^{-1} V^T$$

- Where $\Lambda^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} MM^{-1} &= (V\Lambda V^T)(V\Lambda^{-1} V^T) = V\Lambda(V^T V)\Lambda^{-1} V^T \\ &= V\Lambda\Lambda^{-1} V^T = VV^T = I \end{aligned}$$

- I.e. small eigenvalues become large eigenvalues and vice versa

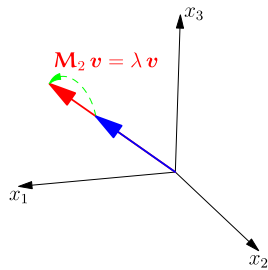
Condition Number

- Taking matrix inverses can be inherently unstable
- Any small error can be amplified by taking the inverse
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)
- For symmetric matrices the largest $|\lambda|$ is the spectral norm, $\|M\|_2$, from the vector spaces lecture
- The condition number is given by

$$\|M\|_2 \times \|M^{-1}\|_2 = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

- Large condition number implies very ill-conditioned

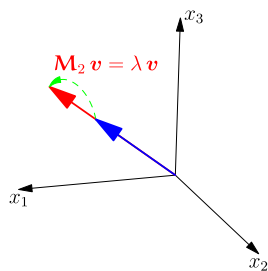
1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



“Inverting” Rank Deficient Matrices

- Rank deficient matrices are non-invertible (i.e. we don't know the vector x such that $Mx = b$) as we don't know the component of the x in the null space
- Although we don't know x we can find a vector, x , that satisfies $Mx = b$ (provided one exists)
- Given a symmetric $n \times n$ matrix with k non-zero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ we can construct a “pseudo-inverse” M^+ as $V\Lambda^+V^T$ where $\Lambda^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$
- This finds the vector x with no component in the null space (it is the solution with the smallest norm)
- This and $(X^TX)^{-1}X^T$ are special cases of the pseudo-inverse we meet with SVD

1. Eigenvectors
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Asymmetric Matrices

- For $n \times n$ real symmetric matrices we are guaranteed there are n orthogonal real eigenvectors
- For non-symmetric matrices the eigenvectors will generally not be orthogonal, they won't necessarily be real
- We are not even guaranteed there will be n linearly independent eigenvectors
- Matrices which don't have n linearly independent eigenvectors are said to be **defective**, e.g. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one
- Randomly generated matrices will typically not be defective, but nevertheless they can arise in certain situations

- The rank of a symmetric matrix, M , is the number of non-zero eigenvalues
- The space spanned by the eigenvectors v_a, v_b , etc. with zero eigenvalue forms a **null space**
- Any vector in the null space will get mapped to the zero vector

$$M(av_a + bv_b + \dots) = \mathbf{0}$$

- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0
- This happens when the columns of the matrix are not linearly independent

Low Rank Approximation

- Recall that matrices with large and small eigenvalues are ill-conditioned so the inverse has the potential to greatly amplify any measurement error
- One work around is to set all small eigenvalues to zero and use the pseudo-inverse
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation
- Low rank approximations are much used to obtain approximate models for arrays of data (we will revisit this when we look at SVD)

Non-Normal Matrices

- Symmetric matrices are part of a class of matrices known as **normal matrices**, which satisfy $MM^\dagger = M^\dagger M$ (for real matrices $M^\dagger = M^T$)
- Hermitian matrices, H , are complex matrices that equal their conjugate transpose: $H = (H^T)^* = H^\dagger$
- Hermitian matrices are also normal; they are important in quantum theory, less so in machine learning
- Normal matrices are exactly those with a full set of orthonormal (possibly complex) eigenvectors
- Most non-symmetric real matrices are not normal (rotations are an exception); their eigenvectors are more confusing

Characteristic Equation

- Since our eigenvectors are defined by $Mv = \lambda v$ then

$$(M - \lambda I)v = \mathbf{0} \text{ has a solution } v \neq \mathbf{0} \text{ only if } \det(M - \lambda I) = 0$$

- This is known as the characteristic equation and leads to an n^{th} order polynomial in λ
- An n^{th} order polynomial has n roots (counting repeats), which are in general complex
- Thus even for real (non-symmetric) matrices we can get complex eigenvalues

Complex Conjugate Pairs

- If $\mathbf{M}$ is real and λ is complex then $\mathbf{v}$ must be complex too, otherwise $\mathbf{M}\mathbf{v}$ would be real but $\lambda\mathbf{v}$ would not
- Furthermore

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} \implies \mathbf{M}\mathbf{v}^* = \lambda^*\mathbf{v}^*$$

- So that eigenvectors and their corresponding eigenvalues exist in complex conjugate pairs

Eigen-Decomposition

- For a non-symmetric matrix, $\mathbf{M}$, provided it is not defective, we can decompose the matrix as

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$$

- $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues
- $\mathbf{V}$ is a matrix whose columns are the right-hand eigenvectors—in general it is not an orthogonal matrix
- $\mathbf{V}^{-1}$ is a matrix whose rows are the left-hand eigenvectors (scaled so that $\mathbf{u}_i^T \mathbf{v}_i = 1$)

Left and Right Eigenvectors

- For symmetric matrices if $\mathbf{v}$ is a right-hand eigenvector in the sense that $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v}^T$ is a left-hand eigenvector

$$\mathbf{v}^T \mathbf{M} = \lambda \mathbf{v}^T$$

- Earlier we used this property to show that the eigenvectors for symmetric matrices are orthogonal
- For asymmetric real matrices the left-hand eigenvectors $\mathbf{u}^T$ are different from the right-hand eigenvectors, although they share the same eigenvalues (they have the same characteristic equation)

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} \quad \mathbf{u}^T \mathbf{M} = \lambda \mathbf{u}^T$$

- As before, $\mathbf{u}_i^T \mathbf{v}_j = 0$ if $\lambda_i \neq \lambda_j$, but in general $\mathbf{v}_i^T \mathbf{v}_j \neq 0$

Summary

- Linear mappings are commonly used in machine learning algorithms such as regression
- We can understand symmetric operators by looking at their eigenvectors
- Any symmetric matrix can be decomposed as $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
 - ★ where $\mathbf{V}$ is an orthogonal matrix whose columns are the eigenvectors
 - ★ and $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues
- This decomposition allows us to understand inverse mappings
- Non-symmetric matrices are more complex—we use the eigen-decomposition less because of this